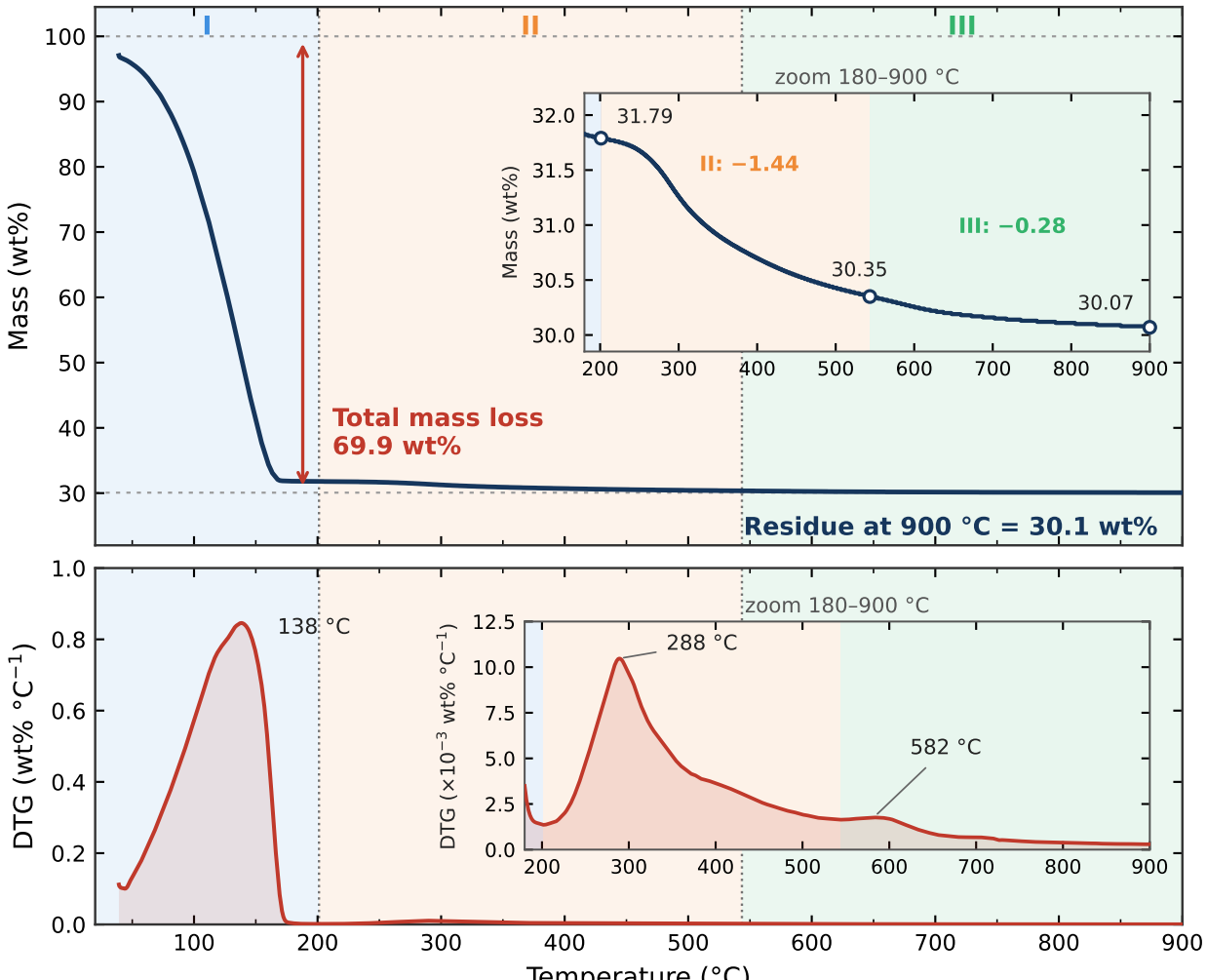
TGA / DTG of silica gel paste – B1 (SG.1)

Paste (wet basis): silica gel 27.91 | bentonite 2.17 | methyl cellulose 0.93 | water 68.99 wt% (dry mix 90/7/3, 20 g + 44.5 g water) • Air, 50 mL/min, 10 °C/min



Air, 50 mL/min, 10 °C/min, 39→900 °C. Step limits at DTG minima (201 and 543 °C).
Mass is relative to the sample loaded in the pan; 2.9 wt% water evaporated before the first data point.

Mass-loss steps

Step	T range	Loss (wt%)	Assignment
I	RT–201 °C	68.2	Free water of the paste + physisorbed water in silica pores and clay [2, 4]
II	201–543 °C	1.44	Burn-off of methyl cellulose [5, 6] + condensation of vicinal silanols [1, 2]
III	543–900 °C	0.28	Dehydroxylation of bentonite [4] + condensation of isolated silanols [1]
Total		69.9	Residue 30.1 wt%: partly dehydroxylated silica + dehydroxylated bentonite

Expected from the paste composition

Step / item	Expected (wt%)	Measured (wt%)
I Water	68.99	68.21
II Methyl cellulose + vicinal OH	0.93 + 0.36	1.44
III Bentonite OH + isolated OH	0.10 + 0.13	0.28
Residue at 900 °C	29.48	30.07

Steps II and III match the methyl cellulose + OH water. Step I is 0.8 wt% lower and the residue 0.6 wt% higher than expected: ≈2 wt% of the paste (water) evaporated between mixing and weighing the TGA sample. OH water from Zhuravlev model [1].

References

[1] Zhuravlev, Colloids Surf. A 173 (2000) 1–38. doi:10.1016/S0927-7757(00)00556-2

[2] Wan, Ramsey, Baran, J. Therm. Anal. Calorim. 99 (2010) 237–243. doi:10.1007/s10973-009-0139-8

[3] Thakkar et al., ACS Appl. Mater. Interfaces 9 (2017) 7489–7498. doi:10.1021/acsami.6b16732

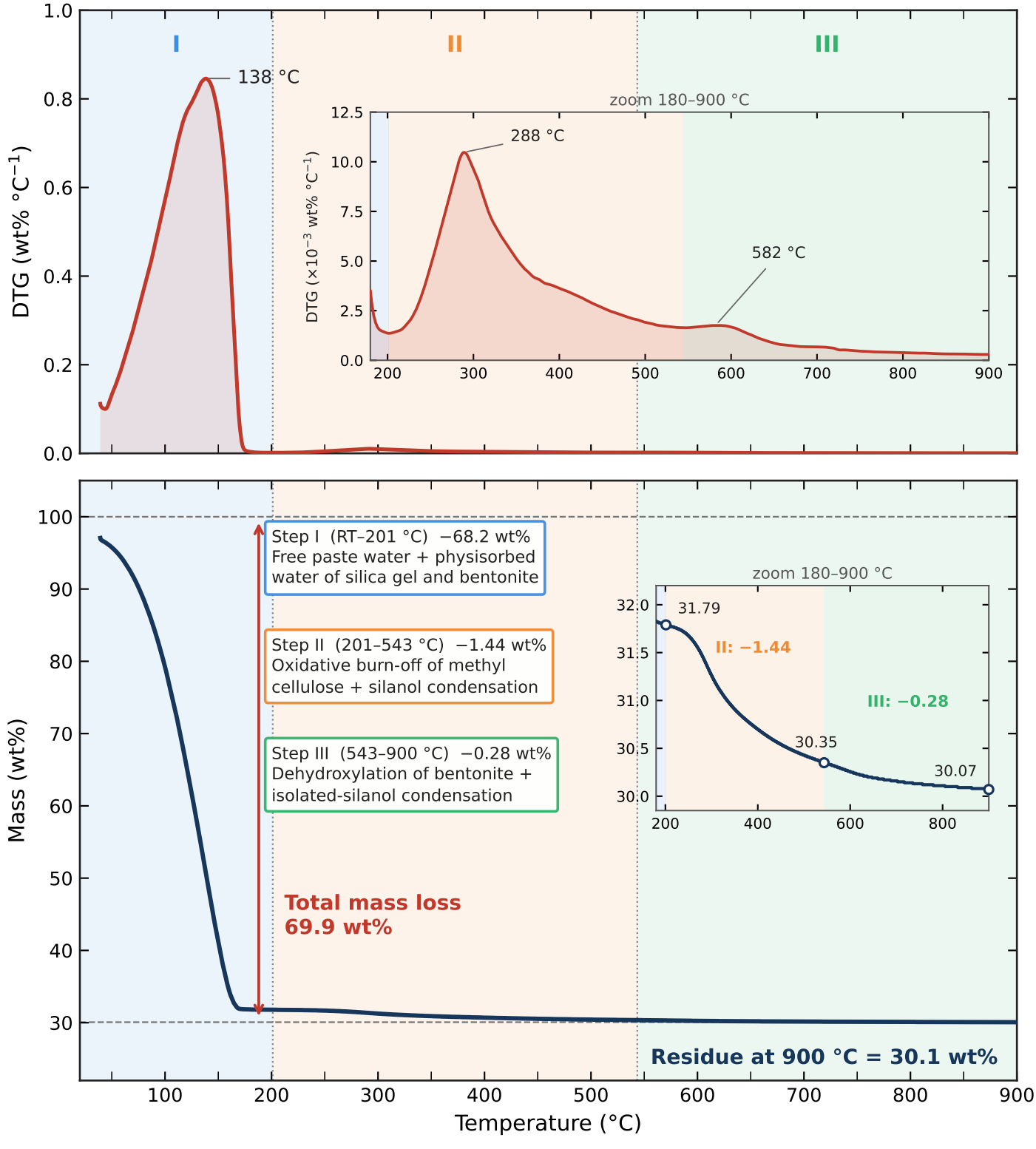
[4] Derkowski & Kuligiewicz, Clays Clay Miner. 70 (2022) 946–972. doi:10.1007/s42860-023-00222-y

[5] Xiao et al., Polymers 13 (2021) 3291. doi:10.3390/polym13193291

[6] Guo et al., Coatings 10 (2020) 499. doi:10.3390/coatings10050499

TGA / DTG of silica gel paste – B1 (SG.1)

silica gel 27.91 | bentonite 2.17 | methyl cellulose 0.93 | water 68.99 wt%
(dry mix 90/7/3, 20 g + 44.5 g water)



Air, 50 mL/min, 10 °C/min, 39→900 °C. Step limits at DTG minima (201 and 543 °C).
Mass is relative to the sample loaded in the pan; 2.9 wt% water evaporated before the first data point.

TGA / DTG Complete Analytical Report – silica gel paste – B1 (SG.1)

Sample ID: Silica Paste B1 • Operator: Shayan • Date: 2026-10-28 • Raw file: Silica Paste B1 SG.1 28102026.xlsx

1. Experimental & Measuring Conditions

Measuring Parameter / Condition	Value
Sample	Silica gel paste, batch B1, sample SG.1 (TGA run 28-10-2026)
Sample ID (vault)	Silica Paste B1
Operator	Shayan
Date	2026-10-28
Raw data file	Silica Paste B1 SG.1 28102026.xlsx
Analysis	Thermogravimetric analysis (TGA) with derivative curve (DTG)
Atmosphere	Air (oxidative)
Gas flow rate	50 mL/min
Heating rate (set)	10 °C/min
Heating rate (fitted from data, 100–800 °C)	9.91 °C/min
Temperature range	39.2 → 900.0 °C
Run time	87.2 min
Number of data points	10464
Sample mass (mg)	Not given in the export – fill in
Mass basis	wt% relative to the mass loaded in the pan (100 %)
First data point	97.06 wt% at 39.2 °C, i.e. 2.94 wt% water evaporated before recording started
Step limits	DTG minima at 201.2 °C and 543.5 °C
Total mass loss (from 100 %)	69.93 wt%
Mass loss in the recorded part only	66.99 wt%
Residue at 900 °C	30.07 wt%
Mass after drying (end of step I)	31.79 wt% = dry paste (silica + bentonite + methyl cellulose + surface OH)
Note on DTG near 720 °C	Small step in the DTG trace at ~720 °C (< 0.0002 wt%/°C) is an instrument artefact, not a reaction

2. Paste Composition (Dry Mix & Wet Paste Basis)

No.	Component	Mass (g)	Dry mix %	Wet paste %	Preparation / Supplier Note
1	Silica gel	18 g	90 %	27.91 %	90 wt% of dry mix; 150 Å pores, 1.15 cm³/g, ~300 m²/g, 35–70 µm
2	Bentonite	1.4 g	7 %	2.17 %	7 wt% of dry mix
3	Methyl cellulose	0.6 g	3 %	0.93 %	3 wt% of dry mix
4	Water	44.5 g	—	68.99 %	Dry mix 90/7/3 (20 g) + 44.5 g water (2.23 g water per g of dry mix)
Total		64.5 g	100 %	100.00 %	

3. Quantitative Mass-Loss Steps Summary

Step	T start	T end	Peak °C	Peak wt%/°C	M start %	M end %	Loss %	Dry-paste %
I	RT (loading)	201.2	138.1	0.846	100.0	31.79	68.21	
II	201.2	543.5	288.1	0.0105	31.79	30.35	1.44	4.53 %
III	543.5	900.0	582.0	0.0018	30.35	30.07	0.28	0.88 %
Total	RT (loading)	900.0			100.0	30.07	69.93	

TGA / DTG Interpretation & Mass Balance – silica gel paste – B1 (SG.1)

Sample ID: Silica Paste B1 • Atmosphere: Air, 50 mL/min, 10 °C/min

4. Step-by-Step Thermal Interpretation

Step	T range	Peak °C	Loss %	Assignment	Detailed Mechanism & Comment	Refs
I	RT–201 °C	138	68.21	Free water of the paste + physisorbed water in the silica gel pores + interlayer water of bentonite	Largest step: one broad DTG peak at 138 °C that ends sharply at ~175 °C, when the last liquid water leaves the 150 Å silica pores. Zhuravlev puts the end of physisorbed-water removal from silica at 190 ± 10 °C; smectite interlayer water also leaves below ~200 °C. After this step the dry paste is 31.79 wt% of the loaded mass.	[1], [2], [4]
II	201–543 °C	288	1.44	Oxidative burn-off of methyl cellulose + condensation of vicinal (hydrogen-bonded) silanols	Small DTG peak at 288 °C with a long tail to ~500 °C. In N2, methyl cellulose has T10 ≈ 292 °C and decomposes at ~343 °C; in air it oxidises at similar or lower temperature, which matches the peak. Vicinal silanols condense (2 Si–OH → Si–O–Si + H2O) mainly between 200 and 450 °C and are removed by ~400 °C; this water gives the long tail. Expressed on the dry paste, step II = 4.53 %.	[1], [2], [5], [6]
III	543–900 °C	582	0.28	Dehydroxylation of bentonite + slow condensation of isolated silanols	Very small loss with a weak DTG peak at 582 °C, typical of the dehydroxylation of montmorillonite (4.7–4.9 % of the clay mass). Isolated silanols condense slowly up to and beyond 900 °C, so the mass is still falling slightly at 900 °C. Expressed on the dry paste, step III = 0.88 %.	[1], [4]

5. Expected from Paste Composition vs. Measured Mass Balance

Component / Balance Item	Expected %	Measured %	Analytical Note
Step I: water	68.99	68.21	Measured includes the 2.9 wt% that left before the first data point (mass is from 100 %).
Step II: methyl cellulose	0.93		3 % of the dry mix.
Step II: vicinal-silanol water (OH 4.6 → ~1.7 nm^{-2}, [1])	0.36		
Step II total	1.29	1.44	Good agreement.
Step III: bentonite OH (4.8 % of clay, [4])	0.1		Upper bound.
Step III: isolated-silanol water (OH ~1.7 → ~0.7 nm^{-2}, [1])	0.13		
Step III total	0.23	0.28	Good agreement.
Silanol water of silica gel (full condensation, % of silica mass)	2.06		From 4.6 OH/nm^2 [1] × 300 m^2/g: upper bound. At 900 °C about 0.5–1 OH/nm^2 remain on silica [1], so only ~80 % of this leaves in the run.
Total mass loss	70.52	69.93	
Residue at 900 °C (silica + bentonite minus their lost OH)	29.48	30.07	The residue is slightly higher and step I slightly lower than expected.
Water lost from the paste before weighing (wt% of paste)		2.0	
Methyl cellulose on dry-paste basis	3.0	4.53	Step II as % of the mass at the end of step I; the extra is mainly vicinal-silanol water.

6. Literature References

[1] Zhuravlev L.T. The surface chemistry of amorphous silica. Zhuravlev model. Colloids and Surfaces A: Physicochemical and Engineering Aspects 173 (2000) 1–38. (doi:10.1016/S0927-7757(00)00556-2)
Used for: Silanol number of amorphous silica 4.6 OH/nm^2; physisorbed water removed by 190 ± 10 °C; vicinal OH condense up to ~400 °C, isolated OH remain to >900 °C

[2] Wan Q., Ramsey C., Baran G. Thermal pretreatment of silica composite filler materials. Journal of Thermal Analysis and Calorimetry 99 (2010) 237–243. (doi:10.1007/s10973-009-0139-8)
Used for: TGA of silica in air: loss up to 200 °C = physically bound water; 200–1000 °C = silanol condensation; dehydroxylation irreversible above 400 °C

[3] Thakkar H., Eastman S., Al-Mamoori A., Hajari A., Rownaghi A.A., Rezaei F. Formulation of aminosilica adsorbents into 3D-printed monoliths and evaluation of their CO2 capture performance. ACS Applied Materials & Interfaces 9 (2017) 7489–7498. (doi:10.1021/acsami.6b16732)
Used for: Background: silica / bentonite / methyl cellulose paste for 3D-printed CO2 adsorbent monoliths

[4] Derkowski A., Kuligiewicz A. Thermal analysis and thermal reactions of smectites: a review of methodology, mechanisms, and kinetics. Clays and Clay Minerals 70 (2022) 946–972. (doi:10.1007/s42860-023-00222-y)
Used for: Smectite (bentonite): dehydration ends below 200 °C; dehydroxylation between ~300 and 900 °C with 4.7–4.9 % mass loss

[5] Xiao Q., Huang M., Zhou X., Dai M., Zhao Z., Zhou H. Effect of molecular weight and degree of substitution on the physical-chemical properties of methylcellulose-starch nanocrystal nanocomposite films. Polymers 13 (2021) 3291. (doi:10.3390/polym13193291)
Used for: Methyl cellulose films in N2 at 10 °C/min: water loss below 100 °C, then chain degradation; T10 = 292 °C

[6] Guo P., Wang F., Duo T., Xiao Z., Xu A., Liu R., Jiang C. Facile fabrication of methylcellulose/PLA membrane with improved properties. Coatings 10 (2020) 499. (doi:10.3390/coatings10050499)
Used for: Methyl cellulose in N2 at 10 °C/min: decomposition temperature 343 °C